

# Internal Components -- Guided Notes ANSWER KEY

Unit tp-2.2 | CompTIA Network+ N10-009 Obj. FC0-U71 2.2 | N10-008 FC0-U71 2.2

For instructor use / distribute after students complete the notes.

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## Question 1

The \_\_\_\_\_ is the main circuit board that connects all other components. It holds the \_\_\_\_\_ socket (where the processor plugs in), RAM slots, PCIe slots, and SATA connectors. The chipset on the motherboard manages \_\_\_\_\_ between the CPU, RAM, and expansion cards.

*Answer: Motherboard; CPU; communication (data flow).*

## Question 2

RAM stands for \_\_\_\_\_ and is a \_\_\_\_\_ (volatile/non-volatile) form of memory, meaning data is lost when power is removed. The CPU uses RAM to store data \_\_\_\_\_ is currently working on. More RAM allows more programs to run \_\_\_\_\_ without performance degradation.

*Answer: Random Access Memory; volatile; it (the CPU); simultaneously.*

## Question 3

The GPU (Graphics Processing Unit) is responsible for rendering \_\_\_\_\_ on a display. It can be \_\_\_\_\_ (built into the motherboard or CPU) or \_\_\_\_\_ (a dedicated add-in card in a PCIe slot). Professional workloads like video editing and machine learning also leverage GPU \_\_\_\_\_ processing power.

*Answer: Images/video (graphics); integrated; discrete (dedicated); parallel.*

## Question 4

The PSU (Power Supply Unit) converts \_\_\_\_\_ current (from the wall outlet) to \_\_\_\_\_ current at voltages the computer components require (typically 3.3V, 5V, and 12V). A PSU is rated in \_\_\_\_\_, which is the maximum total power it can provide to all components at once.

*Answer: Alternating current (AC); direct current (DC); watts.*

## Question 5

NIC stands for \_\_\_\_\_ and provides the hardware interface between the computer and the network. Wired NICs use an \_\_\_\_\_ port (RJ-45 connector). Wireless NICs use \_\_\_\_\_ or \_\_\_\_\_ radio standards. Most modern motherboards have a NIC \_\_\_\_\_ (built in).

*Answer: Network Interface Card; Ethernet; Wi-Fi (802.11); Bluetooth; integrated.*

## Question 6

The CPU cooler sits on top of the CPU and dissipates \_\_\_\_\_ generated by the processor. It transfers heat from the IHS (Integrated Heat Spreader) to either a \_\_\_\_\_ (passive metal fin array) or an \_\_\_\_\_ (liquid loop). Thermal paste between the CPU and cooler improves \_\_\_\_\_ transfer.

*Answer: Heat; heatsink; all-in-one liquid cooler (AIO); thermal (heat).*

### Question 7

PCIe (Peripheral Component Interconnect Express) slots on the motherboard allow expansion cards to be added. Common expansion cards installed in PCIe slots include: \_\_\_\_\_, \_\_\_\_\_, \_\_\_\_\_, and \_\_\_\_\_ (list at least three).

*Answer: Discrete GPU, dedicated NIC, sound card, RAID controller, capture card (any three).*

### Question 8 -- Real World Application

REAL WORLD: A video editor complains their computer is running very slowly when exporting 4K video, even though the system has a fast CPU. Upon investigation, the technician finds only 8 GB of RAM and a motherboard with an integrated GPU. Identify two specific component upgrades that would most improve export performance and explain why each helps.

*Answer: Upgrade 1 -- Add more RAM (to 32 GB or 64 GB): video editing software uses large amounts of RAM to hold frame buffers, preview renders, and project files. With only 8 GB, the OS is likely using the page file (slow SSD/HDD swap), which dramatically reduces performance. More RAM lets the editor work with more data in fast volatile memory. Upgrade 2 -- Add a discrete GPU: 4K video export leverages GPU hardware encoding (NVENC on NVIDIA, AMF on AMD). An integrated GPU lacks dedicated VRAM and compute units, forcing encoding onto the CPU. A dedicated GPU can accelerate encoding by 5-10x for H.264/H.265 export.*